

Chapter 24

Lecture 24: Stone–Čech compactification. Complete metric spaces.

Theorem 24.0.1. [Recall: Stone–Čech compactification] *Let X be a completely regular space. Then there exists a compactification Y of X such that every bounded continuous function $f : X \rightarrow \mathbb{R}$ extends uniquely to a continuous function $\bar{f} : Y \rightarrow \mathbb{R}$. (Y is called the Stone–Čech compactification of X)*

In the last lecture, we have shown the existence part of the above theorem. The uniqueness part is a consequence of the following lemma.

Lemma 24.0.1. *Let $A \subseteq X$. Let $f : A \rightarrow Z$ be continuous and Z be Hausdorff. Then there exists at most one extension of f to a continuous function $g : \bar{A} \rightarrow Z, g|_A = f$.*

Proof

Suppose that $g, g' : \bar{A} \rightarrow Z$ are two different extensions of f . Then there exists $x \in \bar{A}$ such that $g(x) \neq g'(x)$. Since Z is Hausdorff, there exist disjoint open sets U, U' in Z such that $g(x) \in U$ and $g'(x) \in U'$. Choose a neighborhood V of x such that $g(V) \subseteq U$ and $g'(V) \subseteq U'$. Now V intersects A because $x \in \bar{A}$. Take $y \in V \cap A$. Then $g(y) = f(y) = g'(y)$, contradicting the fact that U and U' are disjoint. ■

Theorem 24.0.2. *Let X be completely regular and Y be a compactification of X such that Y satisfies the Stone–Čech property 24.0.1. Then any continuous map $f : X \rightarrow C$ with C being a compact Hausdorff space can be extended uniquely to a continuous map $g : Y \rightarrow C$.*

Proof

C is compact Hausdorff, so it is normal thus is completely regular. So C can be imbedded in a cube $[0, 1]^J$ for some index set J . Since C is homeomorphic to its image in $[0, 1]^J$, we may as well assume C is a subspace of $[0, 1]^J$. Then each component function $f_\alpha = \pi_\alpha \circ f : X \rightarrow [0, 1]$ is a bounded continuous function on X . By the Stone–Čech property, f_α extends to a continuous function $\bar{f}_\alpha : Y \rightarrow [0, 1]$. Define $\bar{f} : Y \rightarrow [0, 1]^J$ by $\bar{f}(y) = (\bar{f}_\alpha(y))_{\alpha \in J}$. Then $\bar{f}$ is continuous and extends f . Now in fact $\bar{f}$ maps Y into C because

$$\bar{f}(Y) = \bar{f}(\bar{X}) \subseteq \overline{\bar{f}(X)} = \overline{f(X)} \subseteq \bar{C} = C. \quad (24.1)$$

where $Y = \bar{X}$ because X is dense in Y , $\bar{f}(\bar{X}) \subseteq \overline{\bar{f}(X)}$ because $\bar{f}$ is continuous, $\overline{\bar{f}(X)} = \overline{f(X)}$ because $\bar{f}$ extends f , and $\bar{C} = C$ because C is closed as a compact subspace of a Hausdorff space. So $\bar{f}$ is a desired extension of f . The uniqueness of the extension follows from the previous lemma. ■

Theorem 24.0.3. *Let Y_1, Y_2 be compactifications of a completely regular space X . Suppose both Y_1 and Y_2 satisfy the Stone–Čech property. Then there exist homeomorphisms $h : Y_1 \rightarrow Y_2$.*

Proof

Consider the inclusion mapping $j_2 : X \rightarrow Y_2$. It is a continuous map of X into the compact Hausdorff space Y_2 . Because Y_1 satisfies the Stone–Čech property, by the previous theorem, j_2 extends to a continuous map $f_2 : Y_1 \rightarrow Y_2$. Similarly, the inclusion mapping $j_1 : X \rightarrow Y_1$ extends to a continuous map $f_1 : Y_2 \rightarrow Y_1$.

The composite $f_1 \circ f_2 : Y_1 \rightarrow Y_1$ has the property that $f_1 \circ f_2(x) = x$ for all $x \in X$. Therefore, $f_1 \circ f_2$ is a continuous extension of the identity map on X . But the identity map of Y_1 is also a continuous extension of the

identity map on X . By the uniqueness of the extension 24.0.1, we have $f_1 \circ f_2 = id_{Y_1}$. Similarly, $f_2 \circ f_1 = id_{Y_2}$. So f_1 and f_2 are homeomorphisms. ■

24.1 Complete Metric Spaces

Definition 24.1.1. Let (X, d) be a metric space. A sequence $\{x_n\} \subseteq X$ is called a *Cauchy sequence* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_n, x_m) < \varepsilon$.

Property 24.1.1. Every convergent sequence is a Cauchy sequence.

Proof Easy exercise.

Definition 24.1.2. A metric space (X, d) is called *complete* if every Cauchy sequence in X converges to a limit in X .

Example 24.1.1 (non-example). $(0, 1)$ with the usual metric. Then the sequence $x_n = \frac{1}{n}$ is a Cauchy sequence that does not converge in $(0, 1)$.

Remark This example also shows that completeness is not a topological property. The subspace $(0, 1)$ of $\mathbb{R}$ is homeomorphic to $\mathbb{R}$, but $(0, 1)$ is not complete while $\mathbb{R}$ is complete.

Example 24.1.2 (non-example). $\mathbb{Q}$ with the usual metric is not complete. For example, the sequence defined by the decimal approximations of $\sqrt{2}$ is a Cauchy sequence in $\mathbb{Q}$ that does not converge to a rational number.

Property 24.1.2. If $A \subseteq X$ is closed and (X, d) is complete, then A is complete in the restricted metric $d|_A$.

Proof Easy exercise.

Property 24.1.3. Let (X, d) be a metric space. Take $\bar{d} = \min\{d, 1\}$. Then (X, d) is complete implies $(X, \bar{d})$ is complete, and conversely.

Proof

A sequence $\{x_n\}$ is Cauchy in d if and only if it is Cauchy in $\bar{d}$. And a sequence $\{x_n\}$ converges under d if and only if it converges under $\bar{d}$. ■

Lemma 24.1.1. A metric space X is complete if and only if every Cauchy sequence has a convergent subsequence.

Proof

“ $\Rightarrow$ ”: Trivial.

“ $\Leftarrow$ ”:

Let $\{x_n\}$ be a Cauchy sequence in (X, d) . We show that if $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$ that converges to some $x \in X$, then $\{x_n\}$ converges to x .

Given $\epsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that for all $m, n \geq N_1$, $d(x_n, x_m) < \frac{\epsilon}{2}$. Since $\{x_{n_k}\}$ converges to x , there exists $N_2 \in \mathbb{N}$ such that for all $n_k \geq N_2$, $d(x_{n_k}, x) < \frac{\epsilon}{2}$. Let $N = \max\{N_1, N_2\}$. Then for all $n, n_k \geq N$, we have

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \quad (24.2)$$

Theorem 24.1.1. $\mathbb{R}^k$ is complete in d_1, d_2, d_∞ .

Proof

We show this for d_∞ .

Let $\{x_n\} \subseteq \mathbb{R}^k$ be a Cauchy sequence in d_∞ . Then $\{x_n\}$ is bounded, i.e., there exists $M > 0$ such that for all n , $d_\infty(x_n, 0) < M$. So such sequence lies in a compact set $[-M, M]^k$. We know in metric spaces, compactness is equivalent to sequential compactness. Thus there exists a convergent subsequence $\{x_{n_k}\}$ that converges to some $x \in [-M, M]^k$. By the lemma, X is complete under d_∞ .

All these metrics are equivalent, so $\mathbb{R}^k$ is complete in d_1, d_2 as well. ■

Lemma 24.1.2. Let $X = \prod_{\alpha \in J} X_\alpha$. Let (x_n) be a sequence in X . Then (x_n) converges to $x \in X$ if and only if for all $\alpha \in J$, the sequence of α -th coordinates $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x) \in X_\alpha$.

Proof

“ $\Rightarrow$ ”: π_α is continuous for all $\alpha \in J$. Then it preserves convergence.

“ $\Leftarrow$ ”: Suppose $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$ for all $\alpha \in J$. Let U be an open neighborhood of x . Then there exists a finite set $K \subseteq J$ and open sets $U_\alpha \subseteq X_\alpha$ for all $\alpha \in K$ such that $x \in \prod_{\alpha \in K} U_\alpha \times \prod_{\alpha \in J \setminus K} X_\alpha \subseteq U$. Since $(\pi_\alpha(x_n))$ converges to $\pi_\alpha(x)$, for each $\alpha \in K$, there exists $N_\alpha \in \mathbb{N}$ such that for all $n \geq N_\alpha$, $\pi_\alpha(x_n) \in U_\alpha$. Let $N = \max_{\alpha \in K} N_\alpha$. Then for all $n \geq N$, $x_n \in U$. Thus (x_n) converges to x . ■

Theorem 24.1.2. *There is a metric on $\mathbb{R}^\omega$ that makes it complete.*

Proof

Define $\bar{d}(a, b) = \min\{1, |a - b|\}$ for all $a, b \in \mathbb{R}$. Define $D(x, y) = \sup_{i \in \mathbb{N}} \{\frac{\bar{d}(x_i, y_i)}{i}\}$ for all $x = (x_i), y = (y_i) \in \mathbb{R}^\omega$. Then D is a metric on $\mathbb{R}^\omega$ that induces the product topology (Please refer to Lecture 15).

Let $\{x_n\} \subseteq \mathbb{R}^\omega$ be a Cauchy sequence in D . Because

$$\bar{d}(\pi_i(x), \pi_i(y)) \leq i \cdot D(x, y) \quad (24.3)$$

we see that for fixed i , $\{\pi_i(x_n)\}$ is a Cauchy sequence in $\mathbb{R}$ under $\bar{d}$ thus is a Cauchy sequence under d . Then it converges, say, to a_i under d . Then the sequence x_n converges to $a = (a_i) \in \mathbb{R}^\omega$ by the previous lemma. ■

24.2 Uniform Metric

Let (Y, d) be a metric space. Let $\bar{d} = \min\{d, 1\}$. We take the set of functions Y^J . There is a metric $\bar{\rho}$ on Y^J defined by $\bar{\rho}(x, y) = \sup_{\alpha \in J} \{\bar{d}(x_\alpha, y_\alpha)\}$. This is a metric (exercise). It is called the uniform metric.

Example 24.2.1. Let $J = [0, 1]$ and $Y = \mathbb{R}$. Define $\bar{d}(f, g) = \sup_{x \in [0, 1]} \{\min(1, |f(x) - g(x)|)\}$.

Theorem 24.2.1. *If Y is complete with respect to d , then Y^J is complete in the uniform metric $\bar{\rho}$ corresponding to d .*

Proof

Recall that if (Y, d) is complete, so is $(Y, \bar{d})$. Now suppose that $f_1, f_2, \dots$ is a Cauchy sequence in Y^J under $\bar{\rho}$. Given $\alpha \in J$, the fact that

$$\bar{d}(f_n(\alpha), f_m(\alpha)) \leq \bar{\rho}(f_n, f_m) \quad (24.4)$$

for all n, m implies that $\{f_n(\alpha)\}$ is a Cauchy sequence in Y under $\bar{d}$. Then it converges to some point y_α . Let f be defined by $f(\alpha) = y_\alpha$. We assert that f_n converges to f in the uniform metric $\bar{\rho}$.

Given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $\bar{\rho}(f_n, f_m) < \frac{\epsilon}{2}$. In particular, given $\alpha \in J$,

$$\bar{d}(f_n(\alpha), f_m(\alpha)) < \frac{\epsilon}{2} \quad (24.5)$$

for all $n, m \geq N$. Letting n and α be fixed and letting $m \rightarrow \infty$ (classical and trivial method in Calculus), we have that

$$\bar{d}(f_n(\alpha), f(\alpha)) \leq \frac{\epsilon}{2} \quad (24.6)$$

This inequality holds for all $\alpha \in J$, provided $n \geq N$. Therefore,

$$\bar{\rho}(f_n, f) = \sup_{\alpha \in J} \{\bar{d}(f_n(\alpha), f(\alpha))\} \leq \frac{\epsilon}{2} < \epsilon \quad (24.7)$$

for all $n \geq N$. So f_n converges to f in the uniform metric $\bar{\rho}$. ■

Remark Recall that the product topology on $\mathbb{R}^J$ is not metrizable if J is uncountable. The topology induced by the uniform metric on $\mathbb{R}^J$ is strictly finer than the product topology so the uniform metric does not metrize the product topology.

The set J here is merely a set, not necessarily a topological space. If J is a topological space, we will have the result in the next theorem. But now we first introduce some definitions.

Definition 24.2.1. Let $f_n : X \rightarrow Y$ be a sequence of functions from the set X to the metric space Y . Let d be the metric on Y . We say that f_n converges uniformly to a function $f : X \rightarrow Y$ if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(f_n(x), f(x)) < \epsilon \quad (24.8)$$

for all $n \geq N$ and all $x \in X$.

Theorem 24.2.2 (Uniform limit theorem). *Let $f_n : X \rightarrow Y$ be a sequence of continuous functions from the topological space X to the metric space Y . If f_n converges uniformly to a function $f : X \rightarrow Y$, then f is continuous.*

Proof This is an easy exercise in the style of ε - δ proof in Calculus. ■

Theorem 24.2.3. *Let X be a topological space and let (Y, d) be a metric space. The set $C(X, Y)$ of continuous functions from X to Y is closed in Y^X under the uniform metric. So is the set $B(X, Y)$ of bounded functions. Therefore, if Y is complete, these spaces are complete in the uniform metric.*

Proof

First, we show that if a sequence of elements f_n of Y^X converges to the element f of Y^X relative to the metric $\bar{\rho}$ on Y^X , then it converges to f uniformly, relative to the metric $\bar{d}$ on Y . Given $\epsilon > 0$, choose an integer N such that

$$\bar{\rho}(f_n, f) < \epsilon \quad (24.9)$$

for all $n \geq N$. Then for all $x \in X$ and all $n \geq N$, we have

$$\bar{d}(f_n(x), f(x)) \leq \bar{\rho}(f_n, f) < \epsilon \quad (24.10)$$

so f_n converges to f uniformly under $\bar{d}$.

Now we show that $C(X, Y)$ is closed in Y^X relative to $\bar{\rho}$. Let f be an element of Y^X that is a limit point of $C(X, Y)$. Then there is a sequence (f_n) of elements of $C(X, Y)$ that converges to f under $\bar{\rho}$. By the uniform limit theorem, f is continuous with $\bar{d}$ as the metric on Y thus f is continuous with d as the metric on Y . So $f \in C(X, Y)$. Therefore, $C(X, Y)$ is closed in Y^X under $\bar{\rho}$.

Finally, we show that $B(X, Y)$ is closed in Y^X under $\bar{\rho}$. Let f be an element of Y^X that is a limit point of $B(X, Y)$. Then there is a sequence (f_n) of elements of $B(X, Y)$ that converges to f under $\bar{\rho}$. There exists N such that $\bar{\rho}(f_n, f) < \frac{1}{2}$. Then, for all $x \in X$, we have $\bar{d}(f_n(x), f(x)) < \frac{1}{2}$, which implies that $d(f_n(x), f(x)) < \frac{1}{2}$. It follows that if M is the diameter of the set $f_N(X)$, then $f(X)$ has diameter at most $M + 1$. So $f(X)$ is bounded. Therefore, $f \in B(X, Y)$. So $B(X, Y)$ is closed in Y^X under $\bar{\rho}$.

By 24.2.1, Y^X is complete under $\bar{\rho}$ if Y is complete under d . Since $C(X, Y)$ and $B(X, Y)$ are closed in Y^X under $\bar{\rho}$, they are complete under $\bar{\rho}$ as well. ■

Definition 24.2.2. *If (Y, d) is a metric space, one can define another metric on the set $B(X, Y)$ of bounded functions from X to Y by the equation*

$$\rho(f, g) = \sup_{x \in X} \{d(f(x), g(x))\} \quad (24.11)$$

It is easy to see that ρ is well-defined, for the set $f(X) \cup g(X)$ is bounded if both $f(X)$ and $g(X)$ are bounded. The metric ρ is called the sup metric.

Remark There is a simple relation between the sup metric and the uniform metric. Indeed, if $f, g \in B(X, Y)$, then

$$\bar{\rho}(f, g) = \min \{\rho(f, g), 1\} \quad (24.12)$$

For if $\rho(f, g) > 1$, then $d(f(x_0), g(x_0)) > 1$ for some $x_0 \in X$. So $\bar{d}(f(x_0), g(x_0)) = 1$ and $\bar{\rho}(f, g) = 1$ by definition. If $\rho(f, g) \leq 1$, then $\bar{d}(f(x), g(x)) = d(f(x), g(x)) \leq 1$ for all $x \in X$. So $\bar{\rho}(f, g) = \sup_{x \in X} \{d(f(x), g(x))\} = \rho(f, g)$.

Remark If X is a compact space, then every continuous function on X is bounded; hence the sup metric is defined on $C(X, Y)$. If Y is complete under d , then $C(X, Y)$ is complete under $\bar{\rho}$, so it is complete under ρ as well.

24.3 Completion

Lemma 24.3.1. *If A is a dense subset of a metric space (Z, ρ) and every Cauchy sequence in A converges to a point in Z , then Z is complete.*

Proof Let (z_n) be a Cauchy sequence in Z . Since A is dense in Z , for each n , there exists $a_n \in A$ such that $\rho(z_n, a_n) < \frac{1}{n}$.

Let $\epsilon > 0$. Since (z_n) is a Cauchy sequence, there exists $N_1 \in \mathbb{N}$ such that for all $m, n \geq N_1$, $\rho(z_n, z_m) < \frac{\epsilon}{3}$. And let $N_2 \in \mathbb{N}$ such that $\frac{1}{N_2} < \frac{\epsilon}{3}$. Let $N = \max\{N_1, N_2\}$. Then for all $m, n \geq N$, we have

$$\rho(a_n, a_m) \leq \rho(a_n, z_n) + \rho(z_n, z_m) + \rho(z_m, a_m) < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \quad (24.13)$$

So (a_n) is a Cauchy sequence in A . By hypothesis, (a_n) converges to some point $z \in Z$. So we have

$$\rho(z_n, z) \leq \rho(z_n, a_n) + \rho(a_n, z) < \frac{1}{n} + \rho(a_n, z) \quad (24.14)$$

for all n . Letting $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} \rho(z_n, z) = 0$. So (z_n) converges to z . Therefore, Z is complete. $\blacksquare$

Theorem 24.3.1. *Let (X, d) be a metric space. There is an isometric imbedding of X into a complete metric space.*

Proof 1

Let x_0 be a fixed point of X . Given $a \in X$, define $\phi_a : X \rightarrow \mathbb{R}$ by the equation

$$\phi_a(x) = d(x, a) - d(x, x_0) \quad (24.15)$$

We assert that ϕ_a is bounded. For it follows, from the inequalities

$$d(x, a) \leq d(x, b) + d(b, a), \quad (24.16)$$

$$d(x, b) \leq d(x, a) + d(a, b), \quad (24.17)$$

that

$$|d(x, a) - d(x, b)| \leq d(a, b). \quad (24.18)$$

Setting $b = x_0$, we have that $|\phi_a(x)| \leq d(a, x_0)$ for all $x \in X$.

Define $\Phi : X \rightarrow B(X, \mathbb{R})$ by setting

$$\Phi(a) = \phi_a \quad (24.19)$$

We show that Φ is an isometric imbedding of (X, d) into the complete metric space $(B(X, \mathbb{R}), \rho)$ (This space is complete by using 24.2.3 and the fact $\mathbb{R}$ is complete). That is, we show that for every pair of points $a, b \in X$,

$$\rho(\phi(a), \phi(b)) = d(a, b). \quad (24.20)$$

By definition,

$$\rho(\phi(a), \phi(b)) = \sup_{x \in X} \{|\phi_a(x) - \phi_b(x)|\} = \sup_{x \in X} \{|d(x, a) - d(x, b)|\} \quad (24.21)$$

We conclude that

$$\rho(\phi(a), \phi(b)) \leq d(a, b) \quad (24.22)$$

for all $a, b \in X$. On the other hand, by setting $x = a$, we have that

$$|d(x, a) - d(x, b)| = d(a, b) \quad (24.23)$$

which implies that $\rho(\phi(a), \phi(b)) \geq d(a, b)$. Therefore, $\rho(\phi(a), \phi(b)) = d(a, b)$ for all $a, b \in X$. So Φ is an isometric imbedding of X into the complete metric space $B(X, \mathbb{R})$. $\blacksquare$

Proof 2

Let $\tilde{X}$ be the set of all Cauchy sequences in X . Define an equivalence relation $\sim$ on $\tilde{X}$ by setting $(x_n) \sim (y_n)$ if and only if $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$.

We assert that $\sim$ is an equivalence relation (easy to verify).

Let Y be the set of all equivalence classes of $\tilde{X}$ under $\sim$. Define a metric D on Y the equation

$$D([(x_n)], [(y_n)]) = \lim_{n \rightarrow \infty} d(x_n, y_n) \quad (24.24)$$

We assert that D is well-defined and is a metric on Y . First we can see that the limit $\lim_{n \rightarrow \infty} d(x_n, y_n)$ exists because $\{d(x_n, y_n)\}$ is a Cauchy sequence in $\mathbb{R}$. Second, if $(x_n) \sim (x'_n)$ and $(y_n) \sim (y'_n)$, then $\lim_{n \rightarrow \infty} d(x_n, x'_n) = 0$ and $\lim_{n \rightarrow \infty} d(y_n, y'_n) = 0$. By the triangle inequality, we have that

$$\lim_{n \rightarrow \infty} d(x'_n, y'_n) \leq \lim_{n \rightarrow \infty} d(x'_n, x_n) + \lim_{n \rightarrow \infty} d(x_n, y_n) + \lim_{n \rightarrow \infty} d(y_n, y'_n) = \lim_{n \rightarrow \infty} d(x_n, y_n) \quad (24.25)$$

and similarly

$$\lim_{n \rightarrow \infty} d(x_n, y_n) \leq \lim_{n \rightarrow \infty} d(x'_n, y'_n) \quad (24.26)$$

thus $\lim_{n \rightarrow \infty} d(x_n, y_n) = \lim_{n \rightarrow \infty} d(x'_n, y'_n)$. So D is well-defined. It is easy to verify that D is a metric on Y .

Define $h : X \rightarrow Y$ by setting $h(x) = [(x, x, x, \dots)]$. Then h is an isometric imbedding of (X, d) into (Y, D) . Indeed, given $x, y \in X$, we have that

$$D(h(x), h(y)) = D([(x, x, x, \dots)], [(y, y, y, \dots)]) = \lim_{n \rightarrow \infty} d(x, y) = d(x, y) \quad (24.27)$$

And h is injective because if $h(x) = h(y)$, then $D(h(x), h(y)) = 0$ thus $d(x, y) = 0 \implies x = y$. An isometry is automatically a homeomorphism onto its image, so h is an isometric imbedding of (X, d) into (Y, D) .

We show that (Y, D) is dense in Y . Let $[(x_n)]$ be an element of Y . We have a sequence $(h(x_n))$ in Y . And we have that

$$D(h(x_n), [(x_n)]) = \lim_{m \rightarrow \infty} d(x_n, x_m) \quad (24.28)$$

So let $\epsilon > 0$ be given. Since $(x_n) \in \tilde{X}$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_n, x_m) < \epsilon$. In particular, for all $n \geq N$, we have that $D(h(x_n), [(x_n)]) = \lim_{m \rightarrow \infty} d(x_n, x_m) < \epsilon$. Hence $(h(x_n))$ converges to $[(x_n)]$. Since $[(x_n)]$ is an arbitrary element of Y , we conclude that $h(X)$ is dense in Y .

Let $(h(x_n))$ be a Cauchy sequence in $h(X)$. Since h is an isometry, (x_n) is a Cauchy sequence in X if and only if $(h(x_n))$ is a Cauchy sequence in $h(X)$. Let $[(y_n)]$ be the element of Y . It follows from the same argument as in the previous paragraph that $(h(y_n))$ converges to $[(y_n)]$. So every Cauchy sequence in $h(X)$ converges to a point in Y .

Since $h(X)$ is dense in Y and every Cauchy sequence in $h(X)$ converges to a point in Y , by the previous lemma, Y is complete. ■

Exercise Let $h : X \rightarrow Y$ and $h' : X \rightarrow Y'$ be isometric imbeddings of the metric space (X, d) in the complete metric spaces (Y, D) and (Y', D') , respectively. Then there is an isometry of $(\overline{h(X)}, D)$ with $(\overline{h'(X)}, D')$ that equals $h' \circ h^{-1}$ on the subspace $h(X)$. In other words, the completion of X is unique up to isometry.